ITAI 1378 Final Project Computer Vision and AI Portfolio

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AI and Robotics Program

Computer Vision

Machine Learning

Python

Google Colab

GitHub

Project Overview

This project portfolio contains computer vision, image processing, and machine learning work completed throughout the semester in ITAI 1378. The projects were developed using Python, Google Colab, Jupyter Notebook, and GitHub. The repository includes CNN experiments, workshop activities, AI agent framework exercises, screenshots, outputs, and supporting documentation demonstrating practical AI and computer vision concepts.

Technologies Used

Python

Google Colab

Jupyter Notebook

GitHub

Computer Vision

Machine Learning

Convolutional Neural Networks

Image Processing

CNN Chihuahua or Muffin Project

This project focused on training a convolutional neural network to classify images as either a Chihuahua or a muffin. The notebook included dataset preparation, image transformations, training loops, loss functions, optimization, and prediction testing using Google Colab and PyTorch.

Image Processing and Computer Vision

Throughout the semester I worked with image processing concepts including resizing, normalization, filtering, transformations, visualization, and image classification workflows. These activities helped build practical understanding of how AI systems process and interpret visual data.

AI Agent Framework Exercises

The agent framework exercises focused on AI workflows, reasoning structures, tool usage, and practical implementation concepts. These notebooks explored how AI agents can process information, follow logic based tasks, and interact with structured workflows using Python.

GitHub Repository Structure

The GitHub repository was organized into separate folders for notebooks, documentation, and results. The repository includes completed notebooks, project documents, screenshots, outputs, AI usage logs, and supporting course materials developed throughout the semester.

Results and Outputs

The repository contains screenshots and notebook outputs showing model training, image classification examples, validation results, and workflow demonstrations from multiple projects completed during the course.

Challenges and Lessons Learned

One of the biggest challenges during the semester was troubleshooting notebook errors, organizing project files, and understanding how machine learning workflows operate step by step. Working through these projects improved my understanding of AI concepts, computer vision systems, and practical problem solving using Python and Google Colab.

Conclusion

This course provided hands on experience with computer vision, machine learning, and AI development workflows. The final portfolio demonstrates practical work completed throughout the semester using notebooks, experiments, documentation, GitHub organization, and AI related project development.